

33 — BAYES THEOREM — POSTERIOR PROBABILITY CALCULATOR

Aim

To calculate posterior probability using Bayes' theorem from given prior probability, likelihood, and evidence.

Software Required

- Python 3.x
- Jupyter Notebook / Google Colab
- Python Standard Library

Procedure

1. Define the prior probability.
2. Define the likelihood.
3. Calculate the evidence probability.
4. Apply Bayes' theorem.
5. Calculate the posterior probability.
6. Display the result.

Code

```
def bayes(prior, likelihood, evidence):  
    return (likelihood * prior) / evidence  
  
p_disease = 0.01  
p_positive_given_disease = 0.99  
p_positive = (  
    p_positive_given_disease * p_disease  
    + 0.05 * (1 - p_disease)  
)  
posterior = bayes(  
    p_disease,  
    p_positive_given_disease,  
    p_positive  
)
```

```
print(  
    f"P(Disease | Positive Test) = {posterior:.4f}"  
)
```

Output:

```
.. P(Disease | Positive Test) = 0.1667
```